Kishanthan Thamilchelvan

Python Full Stack Engineer

sivathamil259@gmail.com +94 076 6011487 kishanthan-thamilchelvan kishanthan100

Professional Summary

Python Full Stack Engineer with 2+ years building full-stack web applications and backend services, from schema and API design to deployment. Python (FastAPI, Flask) on the backend, React/TypeScript on the front end, PostgreSQL and Redis for data, AWS with Docker and Terraform for infrastructure. Secure APIs by default (JWT, RBAC) and ship through CI pipelines that gate on tests.

Experience

Python Full Stack Engineer *Feb 2024 – May 2026*

Sukan Marketing

- Designed and built a customer self-service portal, removing manual sales-rep dependency for order placement
- Built an ETL pipeline to sync ERP data into the reporting database, replacing manual spreadsheet exports
- Built async REST API services with FastAPI to back internal ERP and reporting workflows, choosing async over sync to handle concurrent ERP calls without blocking
- Built an ERP-integrated web application using Flask and PostgreSQL, connecting internal ERP data to customer-facing workflows
- Introduced JWT-based authentication with RBAC authorization across internal services to control access by role
- Introduced automated test coverage into the CI pipeline for core API endpoints, establishing a testing baseline where none existed
- Wrote Bash scripts to automate deployment tasks on EC2, reducing manual deployment steps

Projects

Trados ERP Platform *FastAPI · TypeScript · PostgreSQL · asyncpg · GitHub Actions*

- Built an async REST API with HttpOnly JWT authentication and RBAC-based authorization
- Provisioned AWS infrastructure (VPC, EC2, RDS, S3) via Terraform instead of manual console setup, for reproducible environments
- Containerized the application with Docker and deployed via CI/CD pipeline

ERP Analytics Pipeline *Python · dbt · Apache Airflow · DuckDB*

- Replaced manual reporting with an orchestrated ELT pipeline, using Airflow for scheduled ingestion and task-level failure handling so no manual intervention was needed

Customer Portal *Python · FastAPI . Postgres*

- Connected React/TypeScript frontend to FastAPI backend for real-time order placement
- Added Redis session caching after identifying repeated auth checks as the latency bottleneck, cutting API response time by 38

Technical Skills

- **Backend:** Python, FastAPI, Flask, SQLAlchemy, asyncio, Pydantic, Alembic,Pandas REST API Design
- **Databases:** PostgreSQL, Redis, MongoDB, DuckDB
- **DevOps & Cloud:** Docker, Docker Compose, GitHub Actions, CI/CD Pipelines, AWS (EC2, VPC, RDS, ECR,CloudFront), Terraform, Nginx, Bash/Shell scripting
- **Frontend:** React, TypeScript, Tailwind CSS
- **Data Engineering:** dbt, Apache Airflow, ELT Pipelines
- **Testing:** pytest, coverage-gated CI, test-driven deployment

Education

Bachelor of Engineering Technology in Material Nano Science Technology *2019 – 2023*

Wayamba University of Sri Lanka

Certifications

- Database Design — DataCamp *April 2026 – Jun 2026*
- Intermediate SQL — DataCamp *May 2026– July 2026*